

Chapter 9

THE NERVOUS SYSTEM

The General Nervous System

Introduction – The nervous system forms an elaborate connecting link between the *sense organs*, which respond to various external and internal stimuli, and *effector organs* such as muscles, glands and luminous structures, through which the insect reacts to the stimuli by coordinated behavioural changes (Schmitt, 1962; Roeder, 1963; Smith and Treherne, 1963; Gouin, 1965; Horridge, 1965; Treherne and Beament, 1965; Treherne, 1966, 1974; Hoyle, 1970). In addition, some nerve cells have a special secretory role (p. 273) and the integrity of the nervous system may be required for normal development and regeneration of other tissues (Finlayson, 1960; Nüesch, 1968). The nervous system is composed essentially of ramifying *neuroglial cells* with supporting and nutritive functions (Wigglesworth, 1960; Smith, 1967) and the more numerous *neurons*, which are highly specialized for the relatively rapid generation and conduction of electrochemical nervous impulses (Pipa, 1961; Landolt, 1965; Chiarodo, 1969; Sohal *et al.*, 1972). Processes of each neuron put it into contact by synaptic junctions with other nerve cells (Chiarodo *et al.*, 1970) or with sensory structures or effector organs. One or more of these processes forms a long conducting fibre or *axon* (Narahashi, 1963) ending in a group of fine, branching fibrils; the number of axons enables the neuron to be classified as uni-, bi- or multipolar. Functionally the nerve cells may be sensory (afferent) neurons, conveying impulses inwards from the sense organs (p. 123 ff.), motor or efferent neurons, conveying them outwards to the effector organs, and association neurons (interneurons) that link the other two types. Over most of the nervous system the neurons are grouped together into *ganglia* (p. 102), all with a more or less complex histological structure (Wigglesworth, 1959, and other references cited below). The axons of many neurons are organized into fibre tracts within the ganglion or run out, arranged together, as peripheral nerves (which also contain the incoming axons of afferent neurons). Neuroglial cells occur within and around the ganglia and nerves, forming an outer layer of cells, the perineurium, that secretes an external, more homogeneous, sheath-like neural lamella (Ashurst and Chapman, 1961; Madrell and Treherne, 1967; Lane, 1968, 1972). This contains neutral mucopolysaccharides and collagen fibres and is primarily a supporting structure, while the perineurium helps to form a physiological barrier between the ionic and

other constituents of the blood and those of the nervous tissue (Treherne and Pichon, 1972). The axons are wrapped around by specialized neuroglial cells (Schwann cells) which form a thick myelin sheath in places. Individual axons vary greatly in diameter, the largest forming 'giant axons' adapted for very rapid nervous conduction over considerable distances (Seabrook, 1970b, 1971; Parnas and Dagan, 1971; Harris and Smyth, 1971).

The very large number of neurons (around a million in the brain of *Apis* according to Witthöft, 1967), and the multiplicity of possible connections among them means that the nervous pathways between and within the ganglia are extremely complicated. Some of the coarser ones have long been known from simple histological studies, but more recent mapping of the nerve tracts and individual axonal connections has required a variety of specialized techniques. These include modern silver impregnation methods, electron microscopy, tracing degenerating fibres after surgical transection of peripheral nerves, intraneuronal injection of dye-stuffs and the use of electrophysiological recording after experimental stimulation. The integration of such physiological and histological results is still actively in progress.

It is convenient to divide the insect nervous system into (a) the central nervous system, (b) the visceral nervous system, and (c) the peripheral nervous system, all three parts being connected with each other.

The Central Nervous System

This is the principal division of the nervous system and consists of a double series of ganglia joined by longitudinal and transverse strands of nerve fibres (Fig. 59). The longitudinal cords are termed *connectives* and join a pair of ganglia with those which precede and succeed it. The transverse fibres or *commissures* unite the two ganglia of a pair. Typically there is a pair of ganglia in each segment of the body, but the members of a pair are usually so closely united that they appear as a single ganglion, the commissure being no longer evident externally. The connectives may be separate and distinct throughout the body as in *Machilis* and *Corydalis*, or in the thorax only as in the Orthoptera, Coleoptera and many Lepidopteran larvae, but usually they are so closely approximated as to form a single longitudinal cord. In many cases the ganglia of adjacent segments coalesce to form *ganglionic centres*. Two of these are always present in the head, and varying degrees of coalescence of the thoracic and abdominal ganglia are revealed by a comparative study of the nervous system in different orders of insects.

Transverse sections through a typical ganglion show the outermost perineurium and neural lamella enclosing groups of neurons that form the ganglionic cortex and enclose a central neuropile. The neurons include the usually large, unipolar, motor neurons and the generally smaller interneurons, which may be massed locally into globuli. Some of the larger motor neurons are arranged in bilaterally symmetrical pairs and can be identified from one insect to another (Cohen and Jacklet, 1967). The neuropile is

composed of axons, collaterals and dendrites, the latter often grouped locally into dense glomeruli.

The central nervous system is divisible into the brain or cerebral ganglion, the suboesophageal ganglion and the ventral nerve-cord.

1. **The Brain** (Figs. 60–63) lies just above the oesophagus between the supporting apodemes of the tentorium. It is the dorsal ganglionic centre of the head; its nerve-cells are almost entirely association neurons and it is formed by the coalescence of the first three neuromeres in the embryo. This threefold division is maintained in the completed organ which is divided into corresponding regions which are designated the *protocerebrum*, the *deutocerebrum* and the *tritocerebrum* respectively, though the division is not always apparent externally. Among general works on the structure of the brain are those by Jawlowski (1936), Hanström (1940), Ehnbohm (1948), Goossen (1949), Satija (1957, etc.), Guthrie (1961), Abraham (1967), Groth (1971) and Strausfeld (1976); various more specialized accounts of its component parts are listed below.

The *Protocerebrum* represents the fused ganglia of the acron and the preantennal segment. It forms the greater part of the brain and innervates the compound eyes and ocelli. The protocerebrum is divisible into (a) the protocerebral lobes and (b) the optic lobes.

(a) The *protocerebral lobes* are fused together along the mid line to form a bilobed ganglion. The two lobes are connected by a median commissural system termed the *central body*, towards which fibres converge from various parts of the brain. In addition to the central body there are two smaller commissures, the anterior and posterior dorsal. The *anterior dorsal commissure* passes in front of and above the central body. The *posterior dorsal commissure* is a T-shaped fibre-tract lying behind the anterior commissure. The most conspicuous formations in the neuropile of the protocerebral lobes are the paired *mushroom bodies* or *corpora pedunculata* which are important association-centres (Landolt and Ris, 1966; Mancini and Frontali, 1967; Steiger, 1967; Frontali and Mancini, 1970; Schürmann, 1970, 1972; Pearson, 1971).

Each mushroom body consists essentially of one or two dorsal cup-like *calyces* of neuropile containing masses of neurons, the so-called calyx cells or *globuli cells*. The calyx (or joined calyces) then continue into a stalk or *pedunculus* of nerve fibres which ends deep in the brain by a complicated root system composed of a recurrent *alpha-lobe*, a *beta-lobe* and sometimes even a third *gamma-lobe*. In the Lepidoptera another separate group of globuli cells gives rise to a tract that leads to an additional tripartite *Y-lobe* (Pearson, *loc. cit.*). The axons of the calyx cells send branches to the stalk and the lobes of the root, where they form synapses with fibres from interneurons lying outside the corpora pedunculata. In the region of the brain between the mushroom bodies are four small *ocellar centres* from each of which an ocellar nerve takes its origin. The two outer nerves supply the paired dorsal ocelli, while the two inner nerves unite just outside the brain to form a single nerve supplying the median ocellus.

(b) The *optic lobes* (*optic ganglia* or *optic tract*) form a complex region of the brain and their degree of development is related to that of the compound eyes (e.g. Zuberi, 1963). Their immensely complicated histology was revealed in classical studies by Cajal and Sanchez (1915). This aspect, and the important role of the optic

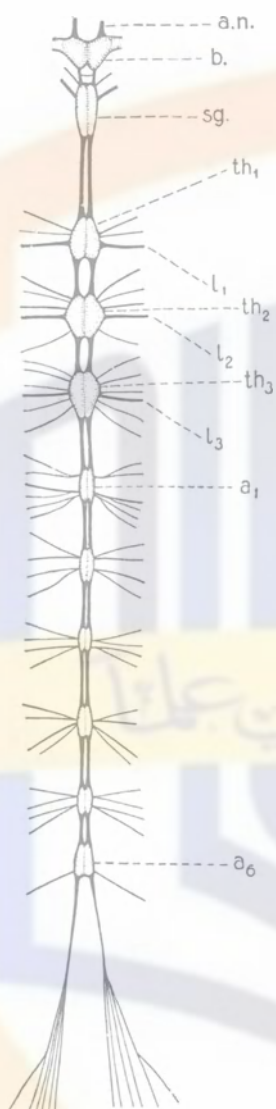


FIG. 59
Central nervous system of
Forficula

a.n., antennal nerve; *b.*, brain;
sg., subesophageal ganglion;
th₁-th₃, thoracic ganglia; *l₁-l₃*,
nerves to legs; *a₁-a₆*, 1st and
terminal abdominal ganglia.

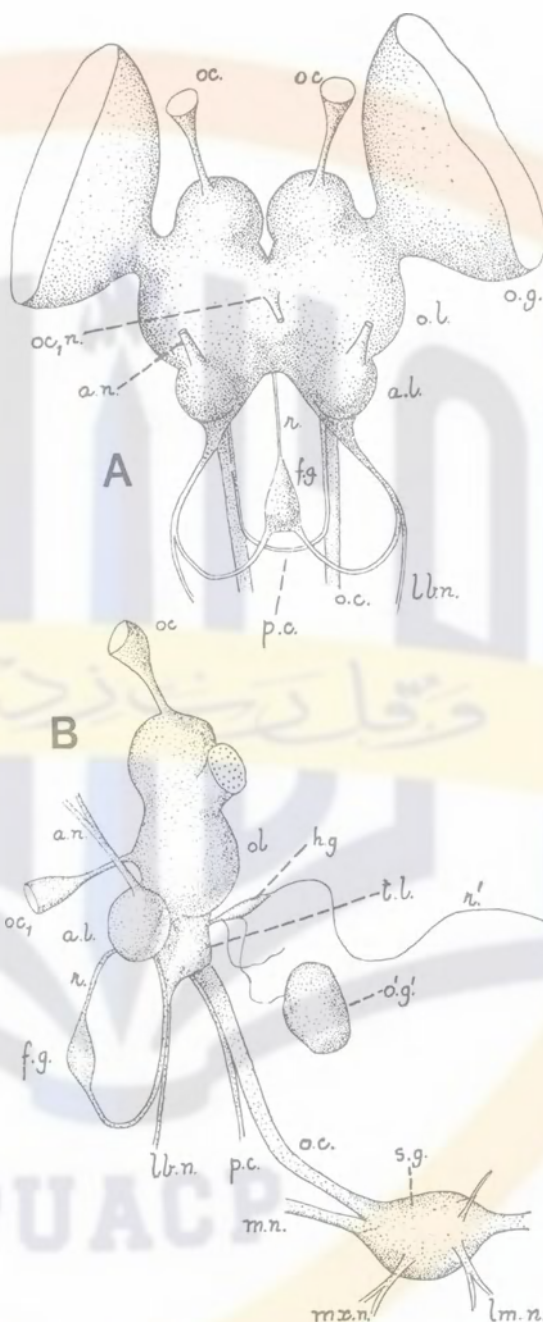


FIG. 60
Brain and subesophageal ganglion of a locust (*Melanoplus*)

A, frontal view; B, lateral view. *a.l.*, antennary lobe; *a.n.*, antennal
nerve; *f.g.*, frontal ganglion; *h.g.*, hypocerebral ganglion; *lb.n.*, labral
nerve; *lm.n.*, labial nerve; *m.n.*, mandibular nerve; *mx.n.*, maxillary
nerve; *oc*, lateral ocellus; *oc₁*, median ocellus; *oc₁n.*, root of nerve to

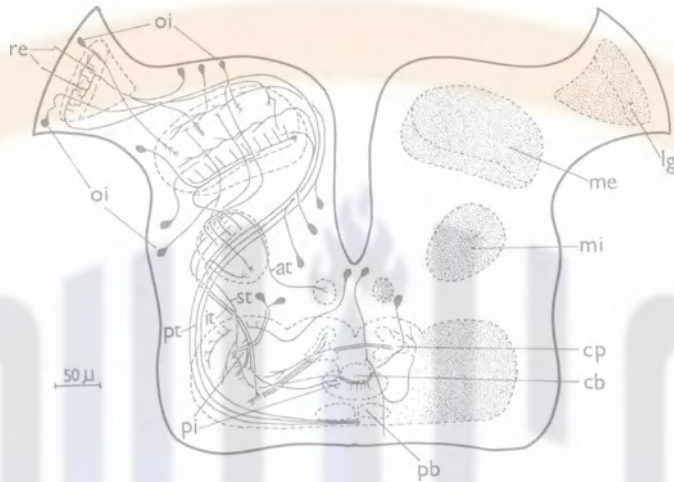


FIG. 61 Diagram to show neurons, nerve-tracts and centres of the protocerebrum and optic lobes of *Gerris lacustris* (after Guthrie, 1962)

at, anterior optic tract; *cb*, central body; *cp*, corpora pedunculata; *it*, inferior median optic tract; *lg*, lamina ganglionaris; *me*, medulla externa; *mi*, medulla interna; *oi*, optic interneurons; *pb*, protocerebral bridge; *pi*, protocerebral interneurons; *pt*, posterior optic tract; *re*, retinal elements; *st*, superior median optic tract.

lobe in processing the sensory input from the compound eye, still requires full elucidation and has attracted many recent workers (e.g. Boschek, 1971; Campos-Ortega and Strausfeld, 1972; Collett, 1970, 1971; Horridge and Meinertzhagen, 1970*a, b*; Strausfeld, 1970, 1971*a, b*; Strausfeld and Blest, 1970; Strausfeld and Braitenberg, 1970; Trujillo-Cenoz, 1965; Varela and Porter, 1969; Varela, 1970; Williams, 1975). Each lobe consists of three principal zones or tracts of nerve tissues which are connected by a similar number of layers of nerve fibres (Figs. 62 and 86). The *lamina ganglionaris* is the zone nearest the eye and is connected with the inner ends of the ommatidia by the layer of *post-retinal fibres*. The middle zone is the *medulla externa* and is connected with the lamina by the *external chiasma* which is formed by the crossing of nerve fibres. The inner zone is the *medulla interna* or *lobula*, united with the preceding zone by the *internal chiasma*. The fibres of the *optic nerve* issue from the inner aspect of the *medulla interna* and divide into anterior and posterior bundles, which pass to the centre of the protocerebrum. Studies of the finer detail show that each major zone includes several types of neurons and a great variety of nervous connections. Short axons from the receptor cells of each ommatidium join those from other ommatidia in the lamina to form a regular array of optic 'cartridges'. The resulting projection of the retinal mosaic on to the lamina is then further projected by other cells into an array of medullary cartridges, while tangential nerve processes intersect the parallel, perpendicularly oriented neurons at various characteristic levels. Links also occur between the two optic lobes and run from the medial part of the protocerebrum to each medulla separately. Electrophysiological techniques have revealed units within the optic lobe mediating different

median ocellus; *o.c.*, para-oesophageal connective; *o.g.*, optic ganglion; *o'.g'*, oesophageal ganglion; *o.l.*, optic lobe; *p.c.*, postoesophageal commissure; *r.*, recurrent nerve (continued in B as the stomatogastric nerve *r'*); *s.g.*, suboesophageal ganglion; *t.l.*, tritocerebral lobe. After Burgess, 2nd Rep. U.S. Ent. Comm.

visual functions: some respond to movement of small parts of the visual field, others react to long, contrasting boundaries moving in a preferred direction; some subtend monocular visual fields, others cover the whole receptive area of both eyes, and so on.

The *Deutocerebrum* represents the fused ganglia of the antennal segment. It is chiefly composed of the paired *antennal* or *olfactory lobes* which are prominent swellings situated on the anteroventral aspect of the brain and innervate the antennae. For their detailed histology and ultrastructure see Schürmann and Wechsler (1969, 1970), Pareto (1972) and Masson (1972).



FIG. 62 Diagram of some major axon pathways in brain of *Periplaneta* showing links from corpora pedunculata (A) and central body (B) to main afferent association areas (after Bullock and Horridge, 1965, and Bretschneider, 1913)

oc, ocellus; 1, second-order fibres from ocellar nerve; 2-4, neurons of optic lobe; 5-6, optic lobe to protocerebral bridge; 8, antennal primary sensory fibres; 9, 11, 12, deutocerebral commissure fibres; 14, antennal motor fibres; 15-17, fibres to calyx of corpora pedunculata; 18, fibres between central body and calyx; 20, globuli neurons of corpora pedunculata.

The so-called *dorsal lobe* is chiefly represented by a transverse fibrous tract situated above the antennal lobes and joining the latter together. Each half of the dorsal lobe is connected with the protocerebral lobe of the opposite side by a chiasma and the antennal lobe is connected with the mushroom body of its side and the central body by the *optico-olfactory chiasma*. Arising from the deutocerebrum are four pairs of nerves: the *antennal nerves* are the longest and most important, and are the sensory nerves of the antennae though the motor fibres mentioned next are also present in the same nerve; each has two roots, one of which is derived from the antennal lobe of its side and the other from the dorsal lobe. The *accessory antennal nerves* issue from the antennal lobes and are the motor nerves of the antennae. The *tegumentary nerves* are a pair of slender strands arising from the dorsal lobe and passing to the vertex.

The *Tritocerebrum* is formed by the ganglia of the third or intercalary segment of the head and is composed mainly of fibres and synapses between the brain, suboesophageal ganglia and frontal ganglion connectives (Willey, 1961). It is divided into two small widely separated lobes which are attached

to the dorsal lobe of the deutocerebrum and receive nerve fibres from the latter. The tritocerebral lobes are joined together by means of the *postoesophageal commissure* which passes immediately behind the oesophagus. They also give origin to (1) the *para-oesophageal connectives* which unite the brain with the suboesophageal ganglion, and (2) the *labro-frontal nerves*. Each of the latter consists of two bundles of fibres, one of which passes to the labrum as the *labral nerve*, and the other forms the root of the frontal ganglion.

2. **The Suboesophageal Ganglion** is the ventral ganglionic centre of the head and is formed by the fusion of the ganglia of the mandibular, maxillary and labial segments. It gives off paired nerves supplying their respective appendages and may also connect with the corpora allata and contribute to the labral nerves (Willey, 1961).

3. **The Ventral Nerve-Cord** consists of a series of ganglia lying on the floor of the thorax and abdomen. They are united into a longitudinal chain by a pair of connectives which issue from the posterior border of the suboesophageal ganglion. The first three ganglia are situated one in each of the thoracic segments, and are known as the thoracic ganglia; the remainder lie in the abdomen and form the ganglia of that region.

The *thoracic ganglia* control the locomotor organs. Each ganglion gives off two pairs of principal nerves, one of which supplies the general musculature of the segment and the other innervates the muscles of the legs. In the meso- and metathorax an additional pair of nerves is present and controls the movements of the wings.

The *abdominal ganglia* are variable in number; in *Machilis* and in many larvae there are eight ganglia in the abdomen but as a rule there are fewer. The first abdominal ganglion frequently coalesces with that of the metathorax and the terminal ganglion is always composite, being formed by the fusion of at least three segmental nerve centres. Each abdominal ganglion gives off a pair of principal nerves to the muscles of its segment.

The histological structure of the ganglia of the ventral chain has been studied by Cohen and Jacklet (1967), Crossman *et al.* (1971), Huddart (1971a), Milburn and Bentley (1971), Pipa, Cook and Richards (1959), Seabrook (1968, 1970a), Young (1969) and others. Fielden (1963) has shown that the roots of the segmental nerves in *Anax imperator* arise from separate dorsal and ventral tracts in the neuropile; sensory activities predominate in the ventral tract and motor activity occurs almost entirely in the dorsal part. Further comparative studies are needed.

The Visceral Nervous System

The visceral or sympathetic nervous system is divided into (1) oesophageal sympathetic, (2) ventral sympathetic, and (3) caudal sympathetic systems.

1. **The Oesophageal Sympathetic (or Stomatogastric) Nervous System** is directly connected with the brain and innervates the fore and middle intestine, heart and certain other parts. It is dorsal in position, lying

above and at the side of the fore intestine and its structure is described in a wide variety of types by Casal (1948), two common arrangements being shown in Fig. 63. Typically, a small triangular *frontal ganglion* lies above the oesophagus, a short distance in front of the brain. The frontal ganglion controls movements of the fore gut (Möhl, 1972). Its removal from immature stages reduces their growth and increases the osmolarity of the blood, apparently through hormonally controlled diuresis (Clark and Anstee, 1971; Penzlin, 1971). Anteriorly the frontal ganglion gives off a *frontal nerve* which passes to the clypeus, and a pair of lateral roots connect the frontal ganglion with the tritocerebrum. Posteriorly it emits a *recurrent nerve* which extends along the mid-dorsal line of the oesophagus and, passing just beneath the brain, expands a short distance behind the latter into a *hypocerebral ganglion*. The recurrent nerve leaves the hypocerebral ganglion as a median or paired lateral oesophageal nerve and passes back to the hinder region of the fore gut, where it (or each branch) terminates in a *ventricular ganglion*. This innervates the adjacent region of the fore and middle intestine. A pair of *corpora cardiaca* lies on the oesophagus just behind the brain, each joined with the hypocerebral ganglion. They are also connected with the protocere-

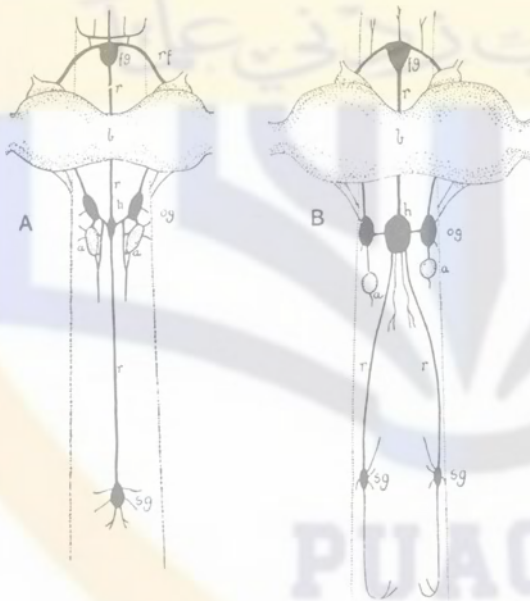


FIG. 63

Semi-diagrammatic figures of two prevalent types of sympathetic nervous system (in black)

A, with a single recurrent nerve and ventricular ganglion; B, with paired recurrent nerves and ganglia. The fore intestine is represented by the dotted lines. *a, a*, corpora allata; *b*, brain; *fg*, frontal ganglion; *h*, hypocerebral ganglion; *og*, corpus cardiacum (right); *r*, recurrent nerve; *rf*, root of frontal ganglion; *sg*, ventricular ganglion.

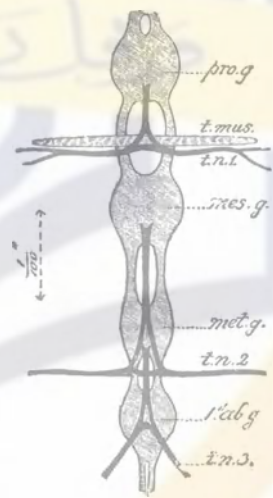


FIG. 64

Thoracic ganglia and portion of sympathetic nervous system of a *Chironomus* larva

pro.g, *mes.g*, *met.g*, thoracic ganglia; *1ab.g*, 1st abdominal ganglion; *t.mus.*, transverse muscle; *tn₁-tn₃*, sympathetic nerves. After Miall and Hammond.

brum, and include both nervous and endocrine structures (p. 274). Connected with them by nerves are the non-nervous *corpora allata*, with important endocrine functions (see p. 276). Among variations on this generalized Pterygote plan may be mentioned the connection of the frontal ganglion with the protocerebrum by a median *nervus connectivus*, the regression of the hypocerebral ganglion and varying degrees of fusion between the hypocerebral ganglion, the corpora cardiaca and the corpora allata, culminating in the condition found in the larvae of Cyclorrhaphan Diptera where all three fuse to form a composite structure known as Weismann's ring which encircles the aorta just behind the brain (p. 278).

2. **The Ventral Sympathetic Nervous System** (Fig. 64), when typically developed (Zawarsin, 1924), consists of a pair of transverse nerves associated with each ganglion of the ventral nerve-cord; each pair is connected with the ganglion preceding it by a median longitudinal nerve. The transverse nerves pass to the spiracles of their segment (Case, 1957; Miller, 1967). In many orders of insects, each transverse nerve is also connected in various ways with the so-called *perisymphetic system*, consisting of neurohaemal organs that release the products of neurosecretory cells in the ventral ganglia (Raabe, 1971).

3. **The so-called Caudal Sympathetic Nervous System** arises from the posterior compound ganglion of the ventral nerve-cord and supplies the reproductive system and posterior part of the gut; it is, however, difficult to draw a clear distinction between such a system and the visceral branches of the peripheral nervous system considered below.

The Peripheral Nervous System

Strictly speaking, this includes all the nerves radiating from the ganglia of the central and sympathetic nervous systems. The ultrastructure of the larger nerves resembles that of ganglia, with an outer lamella of collagen fibres in a mucopolysaccharide matrix secreted by perineurial cells, and with other glial cells lying most abundantly near the larger axons (Osborne, 1966; Huddart, 1971b). Micro-anatomical accounts of peripheral nerve distribution are generally fragmentary, dealing only with the innervation of particular organs but for more detailed descriptions of thoracic and abdominal innervation in selected species see Nüesch (1954, 1957), Wittig (1955), Pipa and Cook (1959), Schmitt (1962) and others. Of morphological interest is their evidence that the nervous system is not very much more conservative in its evolution than are the muscular and exoskeletal systems: muscles which seem to form an integral part of one segment may in part be innervated from the ganglia of other segments.

In some viscera, such as the salivary glands of *Periplaneta*, which are supplied from both the suboesophageal ganglion and the stomatogastric system, nerves reaching an acinus form a plexus on its surface and the axons then penetrate between the acinar cells to end in close contact with the

gland-cell membranes by specially differentiated terminations (Whitehead, 1971). The junctions between peripheral nerve-endings and muscles are also specialized regions (Osborne, 1967), as are the stretch-receptors that form part of the proprioceptor sensory system (p. 130) and from which connections join the central nervous system. Various regions of the gut are elaborately innervated (e.g. Zawarsin, 1916; Orlov, 1924), with multipolar neurons in or on the intestinal wall. In the blowfly *Phormia* this system helps to regulate feeding: a group of 4–8 neurons are located in branches of the median abdominal nerve and respond to changes in the volume of the crop (Gelperin, 1971*a, b*). The reproductive system, too, has a complex innervation concerned in copulation and oviposition (Rehm, 1939; Degrugillier and Leopold, 1972; Grossman and Parnas, 1973). Lastly, the integument is supplied with a delicate plexus of nerve fibres and sensory neurons; these differentiate from epidermal cells and acquire connections distally with cuticular sensilla and proximally with the central nervous system (Zawarsin, 1912*a, b*; Rogosina, 1928; Wigglesworth, 1953; Finlayson, 1972; Hasenfuss, 1973).



FIG. 65 Portion of the peripheral sensory nervous system of a silkworm
h, h, bases of sensory hairs; *s*, bipolar nerve cells; *m*, multipolar nerve cells; *n*, nerves. After Hilton, *Amer. Nat.* 36.

Modifications of the Nervous System

There are many degrees of cerebral development in insects, more or less clearly correlated with the complexity of the insect's sensory equipment or behaviour (Howse, 1975). Thus, the volume of the brain is 1/174th of the body volume in *Apis* and 1/280th in *Formica*, but only 1/3290th in

Melolontha and 1/4200th in *Dytiscus*. Again, the optic lobes are developed in proportion to the size of the eyes and the antennal lobes related to the development of the antennae and the senses mediated by them. Internally, the mushroom bodies attain great size and complexity in Hymenoptera with elaborate behaviour. Structural differences in the brains of drone, worker and queen bees appear to be correlated with the degree of development of the special instincts and activities of the three forms. Further, by comparing the size of the mushroom bodies to the brain as a whole it has been shown in the Hymenoptera that the sawflies come lowest in the scale and the social Hymenoptera highest, while the solitary bees occupy an intermediate position (Hanström, 1926; Pandazis, 1930; Gejvall, 1936; Goossen, 1949; and earlier workers; cf. also Zuberi, 1963, on termites).

In the ventral nerve-cord (Fig. 66), the most generalized condition occurs in the Thysanura, many larvae and some lower Pterygotes where the suboesophageal ganglion, three thoracic and eight abdominal ganglia are separately visible, the most posterior being a composite ganglion. Most Orthopteroid insects, Mecoptera, Trichoptera and Hymenoptera show only a little more concentration than this, but the metathoracic ganglion commonly fuses with

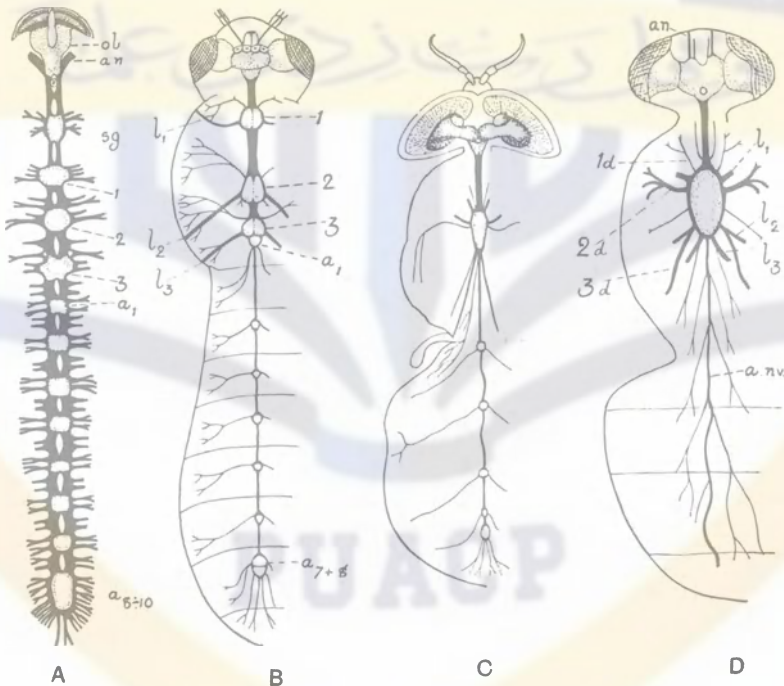


FIG. 66 Schematic figures of the central nervous system showing degrees of concentration, based upon various authors

A, *Machilis* (Oudemans); B, *Chironomus* (Brandt); C, *Stratiomys* (Kunckel d'Herculais); D, *Musca* (Hewitt). 1-3, thoracic ganglia; a₁, a₇₋₈, a₈₋₁₀, abdominal ganglia; an, antennal nerve; ol, optic lobe; sg, suboesophageal ganglion; l₁₋₃, nerves to legs; 1d-3d, dorsal thoracic nerves; a.n.v., abdominal nerves.

the first 1–3 abdominal ones and the 7th and subsequent abdominal ganglia form a compound centre. The other orders show various increasing degrees of fusion, often reaching extremes. Thus, in many Heteroptera, the suboesophageal and prothoracic ganglia are distinct but all the others have fused together. In many Sternorrhynchan Homoptera and higher Diptera, only the suboesophageal ganglion and a single compound thoracico-abdominal ganglion are to be seen while in the Coccoidea, Aphidoidea and some Coleopteran larvae all the ventral ganglia (including the suboesophageal) are united in a single centre. Data amplifying these generalizations will be found in the chapters dealing with individual insect orders.

Physiology of the Nervous System

A detailed treatment of this topic cannot be attempted here and reference should be made to the reviews and selected papers cited on p. 115 ff. The biophysical processes involved in the conduction of nervous impulses in insects seem to be essentially similar to those in other invertebrates (Narahashi, 1963). The resting potential of a giant axon of *Periplaneta* is about 80 mV; when the neuronal membrane is depolarized temporarily (mainly through an increase in its permeability to sodium ions) an action potential of 85–100 mV, lasting about 1 ms, is produced. The passage of such momentary changes of potential along an axon represents the propagation of the nervous impulse. The velocity of conduction varies somewhat; in the giant axons of *Periplaneta* it occurs at about 7 metres per second, while in the afferent nerves from the paraprocts of *Anax* nymphs it is about 2 m s^{-1} (see, for example, Chapman and Pankhurst, 1967).

Synaptic junctions between neurons lie in the ganglionic neuropile (especially the glomeruli), where associations occur between axon branches or dendrites. Interneuronal junctions between cell bodies (Landolt and Ris, 1966) or between axons and somata are rare or absent. The synaptic zones are without glial processes, the participating axon branches or dendrites are separated by a synaptic cleft about 20 nm wide, and the presynaptic knob contains mitochondria, glycogen granules and vesicles. The latter vary from about 30 to 70 nm in diameter with cores that may be electron-dense or electron-transparent (Boistel, 1968; Chiarodo, 1969; Chiarodo *et al.*, 1970; Schürmann, 1970, 1972; and others). In addition to the above chemical synapses, where the vesicles indicate the role of transmitter substances, other inter-neuronal contacts are formed by ephapses or tight junctions between the pre- and post-synaptic membranes; the ultrastructure of such contacts suggests that they are synapses across which electrical transmission occurs (e.g. Steiger, 1967; Huddart, 1971*b*). The identity of the chemical transmitters in the insect nervous system has been discussed extensively (see Pitman, 1971; Gerschenfeld, 1973; McDonald, 1975). Acetylcholine (ACh) and the enzymes that hydrolyse and synthesize it (acetylcholine esterase and choline acetylase) are present in relatively large quantities in insects (Smallman and Mansing, 1969) and it seems likely that acetylcholine is a transmitter at neuropilar axo-axonic synapses; γ -aminobutyric acid (GABA) may also play a part in synaptic transmission in the neuropile. Transmitter substances appear also to be released at neuromuscular junctions: there is evidence that glutamate is responsible for transmitting excitation from nerves to muscle (e.g. Beranek and Miller, 1969) and that GABA may act as an inhibitory transmitter at these sites.

In addition to the more or less complex pathways mapped histologically, several simpler pathways have now been analysed by electrophysiological recording techniques. For example, in *Calliphora* flight starts after a jump due to the contraction of muscles running from the tergum to the trochanter of the mid leg. Mulloney (1969) has shown that these muscles are activated by a motor neuron in the thorax which in turn is driven by two interneurons in the brain, from which descending paths synapse with the motor neurons in the thorax. A more intensively studied pathway is that leading from the cerci or paraproccts through the last abdominal ganglion up ascending giant axons in the ventral nerve cord. In the nymph of the dragonfly *Anax imperator* the giant fibres run the length of the cord (Fielden, 1960); they synapse with efferent fibres in the segmental nerves of the thoracic and abdominal ganglia and a rapid evasion response can be elicited by stimulating the afferent nerves from the paraproccts. In *Periplaneta* the cercal nerve tracts lead directly into the extensively ramifying dendritic processes of the interneurons from which giant axons run forwards through the thorax to activate the antennae and a general alarm system, though the accompanying leg movements are controlled by separate motor neurons with more slowly conducting pathways (Dagan and Parnas, 1970; Milburn and Bentley, 1971; Harris and Smyth, 1971. See also Seabrook, 1970b, 1971, for the giant fibre system of *Schistocerca*).

An important function of the nervous system is the central processing of information derived from the sense organs, especially from the compound eyes and auditory organs, which respond to stimuli with relatively elaborate spatial or temporal patterns. In the ventral nerve cord of *Locusta*, for example, electrophysiological recording has detected several types of symmetrically arranged 'acoustic interneurons' dealing with impulses from the auditory tympanal organs in the first abdominal segment (Kalmring, Rheinlaender and Rehbein, 1972; Kalmring, Rheinlaender and Römer, 1972). Three of these types depend on input from the organ of the same side, another responds to stimuli received by both sides, while a third type is unusual in simply measuring the repetition rate of the sound pattern. Comparable mechanisms have been found in several other insect species (e.g. Fraser Rowell and McKay, 1969; McKay, 1969; Roeder, 1969). The central processes involved in vision are likely to prove much more complicated (Horridge *et al.*, 1965) but it is already clear that electrophysiological units in the optic lobes of selected Orthoptera, Diptera and Lepidoptera respond differentially when a variety of visual stimuli is presented to the insect. Collett (1972), for example, found several directionally sensitive movement detectors in the brain of the hawkmoth *Sphinx ligustri*. These included (a) medial protocerebral neurons with large binocular fields, projecting to the optic lobe or to the ventral nerve cord; (b) movement detectors projecting from the medulla interna of one optic lobe to the opposite medial protocerebrum, and (c) optic lobe output cells which excite the binocular movement detectors centrifugal to the medulla. For other examples see Dingle and Fox (1966), Collett (1970), Mimura (1971) and Northrop and Guignon (1970).

The many interesting and sometimes elaborate forms of instinctive behaviour shown by insects have often been described and some have also been explained as simple taxes and kineses or interpreted in more complex ethological terms (Tinbergen, 1951; Baerends, 1959; Fraenkel and Gunn, 1961; Thorpe, 1963; Haskell, 1966; Evans, 1966; von Frisch, 1967; Sudd, 1967; Ewing and Manning, 1967; Manning, 1972). The neural mechanisms that underlie this behaviour, and especially those by which a simple act is incorporated into a larger biologically

significant behaviour pattern, are now being studied increasingly, despite the many difficulties (Hoyle, 1970). Most attention has been paid to the nervous control of motor activity in respiration, walking, flight, stridulation and mating behaviour. In *Locusta*, for example, there is one neural pacemaker controlling spiracular opening and closing and another centre that controls ventilatory movements of the abdomen (Miller, 1966, 1967). The centres are synchronized by interneurons that run from the ventilation centre to other interneurons which are antecedent to the spiracular motor neurons in the ganglion of the same segment as the spiracle concerned.

Apart from simple evasive and defensive reactions, in which they play a dominant role, reflex arcs seem mainly to be involved in perfecting movements that are initiated by central mechanisms or in helping to maintain the nervous system in a state of adequate excitation. Studies on the nervous control mechanisms of insect flight (Wilson, 1968) suggest that a simple pattern of neural output to the flight-muscles, produced centrally, can support a form of flight, even in the absence of sensory input from wing receptors. Similarly, in a physiological preparation of *Periplaneta* in which all sensory input from the legs had been eliminated, centrally generated patterns of activity could be demonstrated in the motor neurons that control the coxal levator and depressor muscles (Pearson and Iles, 1970). Normal locomotion involves local reflexes, however, as is clearly shown in *Carausius* (Wendler, 1964). Here the proprioceptor hair-plates of the legs lead to a 'negative feedback' loop which moderates the exaggerated steps caused by a separate, centrally generated motor sequence.

The neural mechanisms involved in stridulation and epigamic behaviour of several species of Orthoptera have been intensively studied (Huber, 1965; Elsner and Huber, 1973). By ablation experiments and the local electrical stimulation of parts of the brain of *Gryllus campestris* it has been shown that sites near the mushroom bodies control the production of normal song-rhythms, while stimulation of the central body leads to abnormal songs through the changed temporal pattern of sound-pulses. In the grasshopper *Gomphocerippus rufus* elements of the male courtship pattern can be related to the activities of single motor units and sound production may be described quantitatively by reference to the efferent discharge to known groups of muscles (Elsner, 1968). It is interesting that in *Stenobothrus rubicundus* the same nervous mechanism can induce two different activities: thoracic motor neurons produce identical firing patterns in controlling leg and wing movements so that certain fundamental features of the wing-beat pattern recur in stridulatory movements of the leg (Elsner, 1974).

Some insects are known to show simple forms of learning and memory (Alloway, 1972). They can be trained to run simple mazes or to associate food with colours or other visual stimuli. Bees, ants and some other Aculeate Hymenoptera remember the visual landmarks they use in following routes to and from their nests. The neurophysiological basis of such reactions is unknown, but some analysis of very simple forms of learning has been made (Horridge, 1962; Hoyle, 1965). Headless *Periplaneta* and *Schistocerca* can be trained to flex their legs for long periods in order to avoid an electric shock. The response depends on continuous excitatory discharge to the coxal adductor muscles by only one or two neurons. If a spontaneous fall occurs in this background discharge, a further electric shock causes the frequency to rise again for several minutes or more; conversely, training to reduce the spontaneous discharge may also be accomplished. Simple learning processes of this kind can occur in preparations containing only a single isolated ganglion (Eisenstein, 1972).

Lastly one may note the role played by the nervous system in controlling circadian behavioural patterns. Circadian rhythms show persistent, stable periodicities of about 24 hours and may be observed in developmental, metabolic and behavioural events. Behavioural periodicity of this kind has been found in locomotor, feeding and reproductive activities and are all likely to be controlled through the central nervous system (Brady, 1974). The best studied example is the circadian rhythm of locomotor activity in *Periplaneta*, which is normally more active at night time than in daylight. In order to maintain this circadian rhythm it is only necessary that one protocerebral lobe and the ipsilateral medulla externa and lobula should be in nervous connection with the thorax. The simplest adequate interpretation of the evidence suggests that a driving circadian oscillator in the optic lobe is coupled electrically to the leg muscles via the protocerebrum, nerve cord connectives and thoracic ganglia, the whole system being modulated by sensory inputs which are integrated with the circadian oscillator output in the corpora pedunculata.

Literature on the Nervous System

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